Exp.No:10

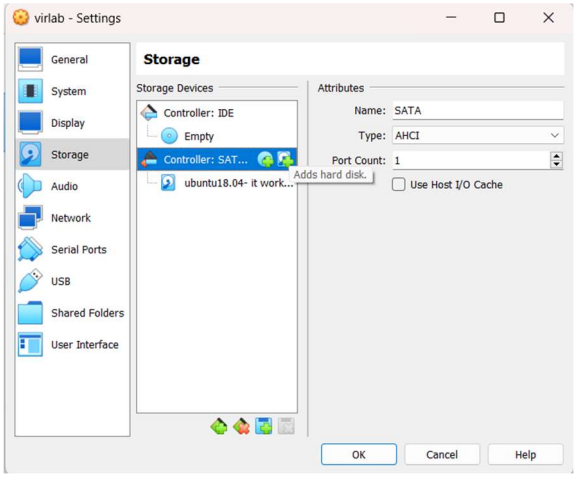
Date:22.09.25 **Creation of VHD on windows and restoring on different host**

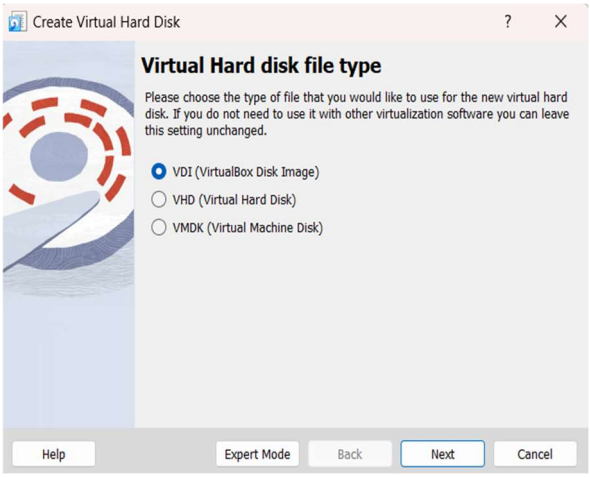
AIM

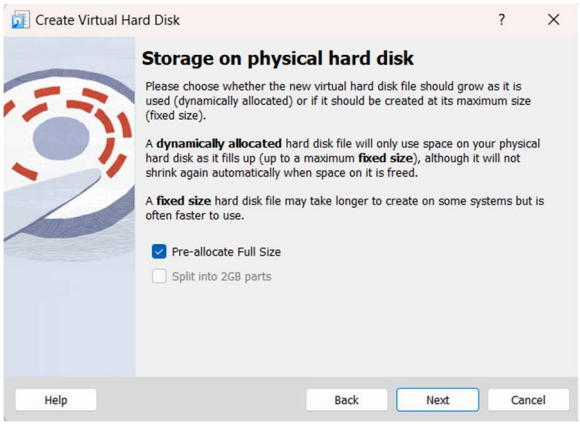
To create a VHD in Windows and restore it on another host system.

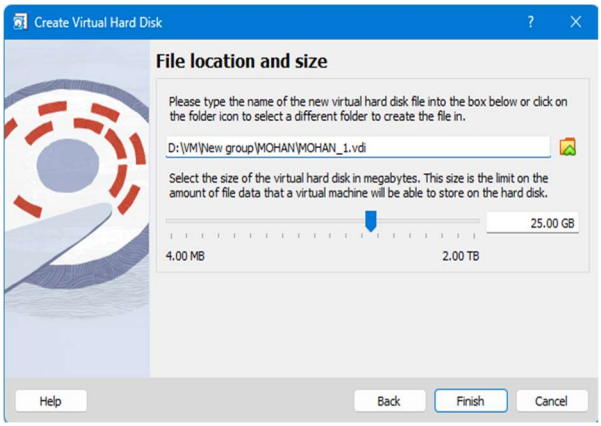
PROCEDURE

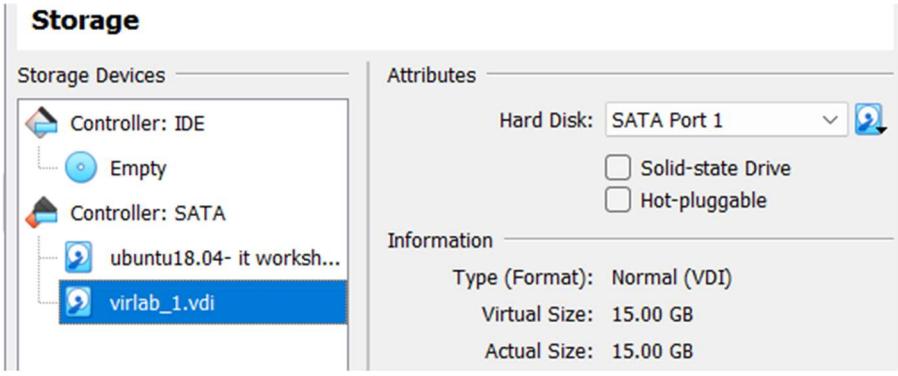
1. Create a Virtual Machine
2. Add Additional Storage to the VM











3. Switch to Root User
4. Check Existing Disks

```
root@tce-VirtualBox:~# fdisk -l
Disk /dev/loop0: 1.7 MiB, 1744896 bytes, 3408 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes


Disk /dev/loop1: 2.1 MiB, 2150400 bytes, 4200 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disk /dev/sda: 50 GiB, 53687091200 bytes, 104857600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd02065ed

Device     Boot Start       End   Sectors Size Id Type
/dev/sda1  *     2048 104855551 104853504  50G 83 Linux


Disk /dev/sdb: 15 GiB, 16106127360 bytes, 31457280 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

5. Initialize the New Disk for LVM

```
root@tce-VirtualBox:~# pvcreate /dev/sdb
  Physical volume "/dev/sdb" successfully created.
```

```
root@tce-VirtualBox:~# pvdisplay
  "/dev/sdb" is a new physical volume of "15.00 GiB"
  --- NEW Physical volume ---
  PV Name               /dev/sdb
  VG Name
  PV Size               15.00 GiB
  Allocatable           NO
  PE Size               0
  Total PE              0
  Free PE               0
  Allocated PE          0
  PV UUID               6UvVus-w0pA-2I3h-nb74-rh39-r6Jf-fakRiY
```

6. Extend the Existing Volume Group

```
root@tce-VirtualBox:~# vgcreate cloud_vg /dev/sdb
  Volume group "cloud_vg" successfully created
```

```
root@tce-VirtualBox:~# vgdisplay
  --- Volume group ---
  VG Name               cloud_vg
  System ID
  Format                lvm2
  Metadata Areas        1
  Metadata Sequence No  1
  VG Access             read/write
  VG Status             resizable
  MAX LV                0
  Cur LV                0
  Open LV               0
  Max PV                0
  Cur PV                1
  Act PV                1
  VG Size               <15.00 GiB
  PE Size               4.00 MiB
  Total PE              3839
  Alloc PE / Size       0 / 0
  Free  PE / Size       3839 / <15.00 GiB
  VG UUID               afvQNR-rVZY-M1HD-JhLG-5o9q-jGOa-Dvi8xy
```

7. Create a New Logical Volume

```
root@tce-VirtualBox:~# sudo lvcreate -n root -L 10G cloud_vg
  Logical volume "root" created.
```

```
root@tce-VirtualBox:~# lvdisplay
  --- Logical volume ---
  LV Path                /dev/cloud_vg/root
  LV Name                root
  VG Name                cloud_vg
  LV UUID                aIYKO3-M1tI-ImGr-dGH1-L8nD-xF2y-RfiK5H
  LV Write Access        read/write
  LV Creation host, time tce-VirtualBox, 2025-09-26 06:27:14 +0530
  LV Status              available
  # open                 0
  LV Size                10.00 GiB
  Current LE             2560
  Segments               1
  Allocation             inherit
  Read ahead sectors     auto
  - currently set to     256
  Block device           253:0
```

8. Extend Logical Volume

```
root@tce-VirtualBox:~# lvextend -L +2G /dev/cloud_vg/root
  Size of logical volume cloud_vg/root changed from 10.00 GiB (2560 extents) to
 12.00 GiB (3072 extents).
  Logical volume cloud_vg/root successfully resized.
```

```
root@tce-VirtualBox:~# lvdisplay
  --- Logical volume ---
  LV Path                /dev/cloud_vg/root
  LV Name                root
  VG Name                cloud_vg
  LV UUID                aIYKO3-M1tI-ImGr-dGH1-L8nD-xF2y-RfiK5H
  LV Write Access        read/write
  LV Creation host, time tce-VirtualBox, 2025-09-26 06:27:14 +0530
  LV Status              available
  # open                 0
  LV Size                12.00 GiB
  Current LE             3072
  Segments               1
  Allocation             inherit
  Read ahead sectors     auto
  - currently set to     256
  Block device           253:0
```

9. Verify Disk Space

```
root@tce-VirtualBox:~# df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            970M     0  970M   0% /dev
tmpfs           199M  1.6M  198M   1% /run
/dev/sda1        49G   11G   36G  23% /
tmpfs           994M     0  994M   0% /dev/shm
tmpfs           5.0M  4.0K  5.0M   1% /run/lock
tmpfs           994M     0  994M   0% /sys/fs/cgroup
/dev/loop1      2.2M  2.2M     0 100% /snap/gnome-calculator/972
/dev/loop0      1.8M  1.8M     0 100% /snap/gnome-system-monitor/193
/dev/loop3       82M   82M     0 100% /snap/gtk-common-themes/1534
/dev/loop4      768K  768K     0 100% /snap/gnome-characters/741
/dev/loop5      2.7M  2.7M     0 100% /snap/gnome-calculator/920
/dev/loop6       92M   92M     0 100% /snap/gtk-common-themes/1535
/dev/loop7       64M   64M     0 100% /snap/core20/2599
/dev/loop8      350M  350M     0 100% /snap/gnome-3-38-2004/143
/dev/loop9      255M  255M     0 100% /snap/gnome-3-38-2004/106
/dev/loop10     896K  896K     0 100% /snap/gnome-logs/129
/dev/loop12     219M  219M     0 100% /snap/gnome-3-34-1804/93
/dev/loop13     219M  219M     0 100% /snap/gnome-3-34-1804/77
/dev/loop14      56M   56M     0 100% /snap/core18/2940
/dev/loop15     2.7M  2.7M     0 100% /snap/gnome-system-monitor/174
```

10. Resize Filesystem

```
root@tce-VirtualBox:~# sudo mkfs.ext4 /dev/cloud_vg/root
mke2fs 1.44.1 (24-Mar-2018)
Creating filesystem with 3145728 4k blocks and 786432 inodes
Filesystem UUID: 8c0f581b-a495-46ea-bc16-5b5cb1c8573d
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done
```

```
root@tce-VirtualBox:~# sudo mkdir -p /mnt/data
```

```
root@tce-VirtualBox:~# sudo mount /dev/cloud_vg/root /mnt/data
```

11. Verify Disk Space after resizing

```
root@tce-VirtualBox:~# df -h
Filesystem                 Size  Used Avail Use% Mounted on
udev                       969M     0  969M   0% /dev
tmpfs                      199M  1.6M  198M   1% /run
/dev/sda1                   49G   11G   36G  24% /
tmpfs                      994M     0  994M   0% /dev/shm
tmpfs                      5.0M  4.0K  5.0M   1% /run/lock
tmpfs                      994M     0  994M   0% /sys/fs/cgroup
/dev/loop0                 219M  219M     0 100% /snap/gnome-3-34-1804/77
/dev/loop10                128K  128K     0 100% /snap/bare/5
/dev/loop8                 768K  768K     0 100% /snap/gnome-characters/741
/dev/loop2                 2.2M  2.2M     0 100% /snap/gnome-calculator/972
/dev/loop6                 9.5M  9.5M     0 100% /snap/gnome-characters/805
/dev/loop9                 896K  896K     0 100% /snap/gnome-logs/129
/dev/loop3                  67M   67M     0 100% /snap/core24/1151
/dev/loop1                 219M  219M     0 100% /snap/gnome-3-34-1804/93
/dev/loop4                  51M   51M     0 100% /snap/snapd/25202
/dev/loop11                 92M   92M     0 100% /snap/gtk-common-themes/1535
/dev/loop7                  82M   82M     0 100% /snap/gtk-common-themes/1534
/dev/loop5                 255M  255M     0 100% /snap/gnome-3-38-2004/106
/dev/loop12                 56M   56M     0 100% /snap/core18/2940
/dev/loop13                 64M   64M     0 100% /snap/core20/2599
/dev/loop14                1.8M  1.8M     0 100% /snap/gnome-system-monitor/193
/dev/loop15                640K  640K     0 100% /snap/gnome-logs/112
/dev/loop16                 56M   56M     0 100% /snap/core18/2952
/dev/loop17                 64M   64M     0 100% /snap/core20/2669
/dev/loop18                350M  350M     0 100% /snap/gnome-3-38-2004/143
/dev/loop19                2.7M  2.7M     0 100% /snap/gnome-calculator/920
/dev/loop20                291M  291M     0 100% /snap/mesa-2404/912
/dev/loop21                2.7M  2.7M     0 100% /snap/gnome-system-monitor/174
/dev/loop22                619M  619M     0 100% /snap/gnome-46-2404/125
/dev/loop23                 67M   67M     0 100% /snap/core24/1055
tmpfs                      199M   28K  199M   1% /run/user/121
tmpfs                      199M   40K  199M   1% /run/user/1000
/dev/mapper/cloud_vg-root   12G   41M   12G   1% /mnt/data
```

RESULT

Thus, the VHD was successfully created on the Windows host and restored on a different host, making the virtual disk accessible and ready for use.